Discussion: Hedge Fund Performance and the U.S.–China Tension

Author: Xiaoyu Zhao (UMass Amherst)

Discussant: Junyong Kim (UNC Pembroke)

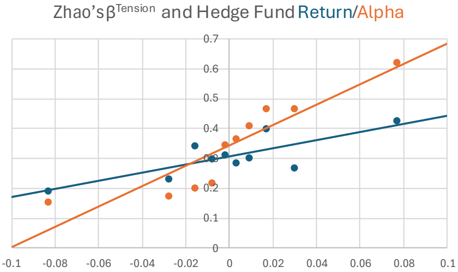
2026 Academy of Finance Annual Meeting

Summary

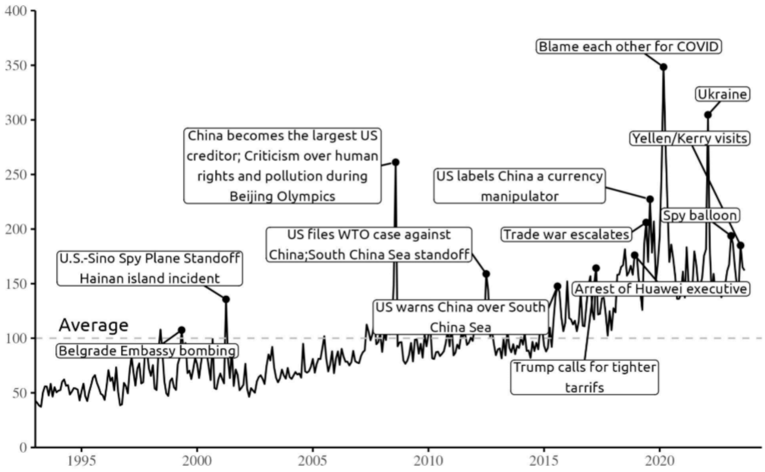
- Hedge fund return space positively prices US-China tension
  - Rogers et al. (2024 SSRN)
  - $\beta_{it}^{Tension}$ in equation (2)
- These hedge funds are able to actively time the US-China tension
  - Liquidity (Cao et al., 2013 JFE), economic policy uncertainty (Liang et al., 2021 SSRN)
  - θ_i in equation (4)
- Personal thoughts
  - Elevated US-China tension is influencing financial markets as well as the economy, making this paper timely and motivated
  - If the tension must be the priced factor, then why should it be hedge funds only? What if mutual funds or stocks more generally?

Zhao (2026 AoF) Table 5

Decile	β^{Tension}	Return (%)	Alpha (FH11)	MFee (%)	IFee (%)	HWM (Dummy)	Lockup (Yrs)	Lev. (Dummy)	Redem. (Days)
1	−0.083	0.194	0.154	1.419	16.688	0.842	1.196	0.632	32.622
2	−0.028	0.233	0.175	1.392	16.738	0.826	1.265	0.621	33.985
3	−0.016	0.343	0.203	1.380	16.380	0.818	1.213	0.608	33.623
4	−0.008	0.301	0.218	1.374	16.500	0.832	1.251	0.608	34.527
5	−0.002	0.313	0.348	1.360	16.328	0.820	1.216	0.607	33.710
6	0.003	0.287	0.367	1.367	16.432	0.815	1.186	0.614	33.815
7	0.009	0.304	0.413	1.373	16.679	0.828	1.169	0.616	34.314
8	0.017	0.403	0.469	1.384	16.631	0.824	1.169	0.623	34.117
9	0.030	0.269	0.470	1.397	16.876	0.833	1.236	0.636	34.687
10	0.077	0.430	0.625	1.410	16.420	0.811	1.053	0.639	31.975



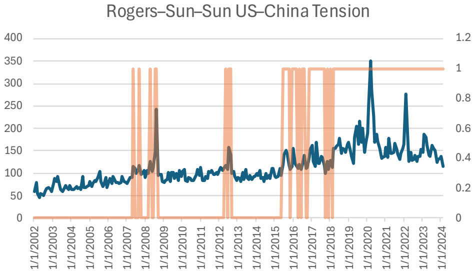
Rogers et al. (2024 SSRN) Figure 1

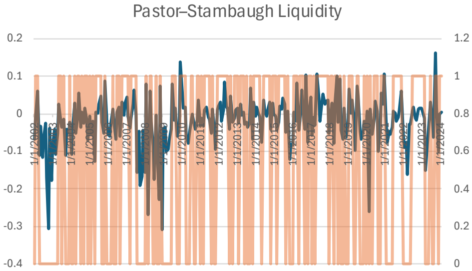


Comments

- What if difference ($\Delta\mathrm{Tension}$) → orthogonalize ($\Delta\mathrm{Tension}^{\perp}$) instead of orthogonalize ($\mathrm{Tension}^{\perp}$) → difference ($\Delta\mathrm{Tension}^{\perp}$)?
  - Many finance papers orthogonalize changes/returns rather than levels/indices: Cao et al. (2020 JFM), Harvey and Liu (2021 JFE), Pukthuanthong et al. (2022 SSRN), Arnott et al. (2023 RFS) (cf. Baker and Wurgler, 2006 JF, Liang et al., 2021 SSRN)
- Equation 4 adopts UCT_t (level); what if $\Delta\mathrm{Tension}^{\perp}$ (change)?
  - Or what if $\mathrm{Tension}^{\perp}$ (level) (ex. $\mathrm{EPU}_t^{\perp}$ in Liang et al., 2021 SSRN equation 7)?
  - Because equation 5 relates β^{Tension} (estimated from $\Delta\mathrm{Tension}^{\perp}$) and θ (estimated from UCT_t)
  - Ex. Treynor–Mazuy: $R_p^e = \alpha_p + \beta_p R_m^e + \gamma_p R_m^e R_m^e + \varepsilon_p$ (change)
  - Cao et al. (2013 JFE, 2020 JFM): L_m captures change rather than level?
- Table 7 runs panel regressions; what if Fama–MacBeth (Chen et al., 2021 JF, Liang et al., 2021 SSRN)?
  - Regressors other than Size and Redemption Periods (Days) are insignificant?

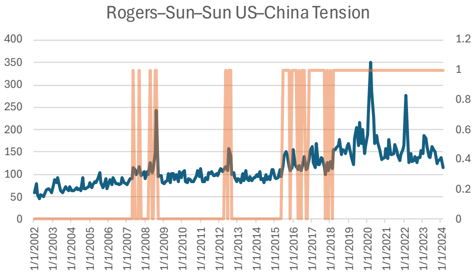
Rogers–Sun–Sun Versus Pastor–Stambaugh

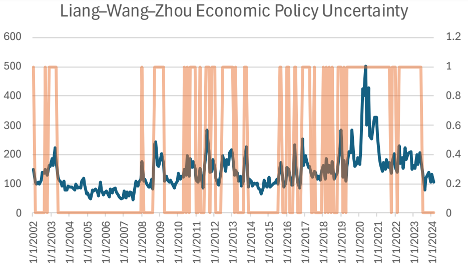


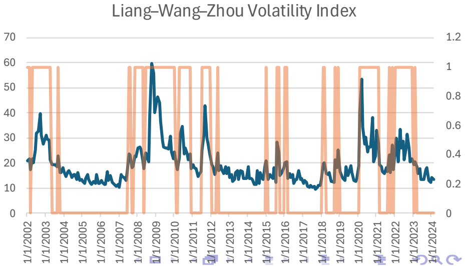


The ones in the dummy time series are evenly distributed throughout the entire sample.

Rogers–Sun–Sun Versus Liang–Wang–Zhou







Chen et al. (2021 JF) Table 4

Table IV

Cross-Sectional Regressions of Fund Performance on Sentiment Beta

This table reports results from Fama-MacBeth (1973) cross-sectional regressions of hedge fund excess return, as well as alpha, on sentiment beta, controlling for fund characteristics and style dummies. In each month and for each hedge fund with at least 30 return observations over the past 36 months, sentiment beta is estimated by regressing the fund excess returns on the Baker-Wurgler (2007) sentiment changes index with controls for the Fung-Hsieh (2004) seven factors (including market excess returns, a size factor, ΔTerm, ΔCredit, and three trend-following factors on bonds, currencies, and commodities), the momentum factor, the Pastor-Stambaugh (2003) liquidity factor, the inflation rate, and the default spread. Then, we perform cross-sectional regressions of fund excess return, or alpha, over the next month on sentiment beta with controls for fund characteristics and style dummies. The fund characteristics include fund size, fund age, management fee, incentive fee, a high-water mark dummy equal to 1 if a high-water mark provision is used, and 0 otherwise, lockup period, and redemption notice period. Both monthly excess return and alpha are reported in percentage. t-Statistics are based on Newey-West (1987) standard errors with two lags.

	Dependent Variable							
	Excess Return				Alpha			
	Coeff.	t-Stat	Coeff.	t-Stat	Coeff.	t-Stat	Coeff.	t-Stat
Sentiment beta	0.17	3.03	0.16	2.84	0.14	3.31	0.12	3.16
Log(fund size)			0.01	0.12			0.01	0.21

Miscellaneous

- Figure 1: Jan 2002–Dec 2024 or Feb 2024 (EPU Rogers)?
- Summary statistics for Tension (UCT), $\text{Tension}^{\perp}$, and $\Delta\text{Tension}^{\perp}$, in addition to Figure 2 and Table 3
- t-statistics (or standard errors) for the αs in Table 5
- In Table 5, $\bar{R}_{10} - \bar{R}_1 = .236\%$ ($\hat{a}_{10} - \hat{a}_1 = .471\%$); in Table 6, they are .180% and .331%, respectively
- In page 16, "... a greater share of funds reduce market exposure to the market during periods ..."
- In equation 5, is High $\text{Tension}_t = 1\{\text{UCT}_t > \overline{\text{UCT}}\}$?
  - Consider using consistent notation (either UCT or Tension) throughout
  - In page 20, "... from equation (4). HighTension_t is an indicator ..."

Cao et al. (2013 JFE) Table 4

For K = 3, $R_1 - R_{10}$ = .513% − .180% = .333%

	All funds				Hedge funds			
Portfolio	$K=3$	6	9	12	$K=3$	6	9	12
Portfolio 1 (top timers)	0.513	0.522	0.525	0.508	0.620	0.625	0.630	0.608
	(2.96)	(3.26)	(3.46)	(3.48)	(3.34)	(3.53)	(3.75)	(3.73)
Portfolio 2	0.323	0.322	0.310	0.297	0.348	0.354	0.342	0.337
	(3.18)	(3.32)	(3.18)	(3.05)	(2.99)	(3.38)	(3.56)	(3.50)
Portfolio 3	0.257	0.259	0.256	0.249	0.360	0.339	0.333	0.319
	(3.26)	(3.44)	(3.27)	(3.11)	(4.36)	(4.47)	(4.02)	(3.93)
Portfolio 4	0.228	0.239	0.232	0.222	0.302	0.312	0.301	0.292
	(2.29)	(2.96)	(2.92)	(2.65)	(3.72)	(4.41)	(4.22)	(3.95)
Portfolio 5	0.203	0.214	0.216	0.211	0.271	0.292	0.294	0.292
	(2.62)	(2.61)	(2.63)	(2.55)	(4.07)	(4.37)	(4.35)	(3.92)
Portfolio 6	0.195	0.197	0.202	0.198	0.240	0.240	0.247	0.249
	(1.98)	(1.96)	(2.07)	(2.09)	(2.48)	(2.52)	(2.75)	(2.88)
Portfolio 7	0.166	0.172	0.171	0.195	0.169	0.220	0.232	0.247
	(1.68)	(1.75)	(1.71)	(1.90)	(1.94)	(2.36)	(2.51)	(2.67)
Portfolio 8	0.171	0.182	0.197	0.209	0.308	0.294	0.295	0.307
	(1.22)	(1.48)	(1.59)	(1.67)	(2.10)	(2.20)	(2.20)	(2.32)
Portfolio 9	0.155	0.145	0.163	0.171	0.226	0.213	0.233	0.240
	(1.34)	(1.05)	(1.26)	(1.32)	(2.10)	(1.52)	(1.57)	(1.58)
Portfolio 10 (bottom timers)	0.180	0.102	0.070	0.081	0.255	0.192	0.162	0.176
	(1.01)	(0.54)	(0.38)	(0.46)	(1.21)	(0.89)	(0.76)	(0.83)
Spread (Portfolio 1–Portfolio 10)	0.333	0.420	0.455	0.427	0.365	0.433	0.468	0.432
	(1.79)	(2.33)	(2.63)	(2.51)	(1.65)	(2.01)	(2.28)	(2.13)

Chen et al. (2021 JF) Table 3

$\alpha_{10} - \alpha_1 = .51\% - (-.08\%) = .59\%$

Portfolio	Sentiment Beta	Excess Return	*t*-Stat	Alpha	*t*-Stat
1 (Low)	–1.07	0.27	1.77	–0.08	–0.51
2	–0.45	0.25	1.94	0.07	0.59
3	–0.25	0.34	2.95	0.14	1.48
4	–0.14	0.30	2.85	0.14	1.34
5	–0.06	0.25	2.35	0.04	0.36
6	0.02	0.25	2.51	0.10	0.96
7	0.10	0.30	2.76	0.12	1.15
8	0.21	0.34	2.99	0.21	1.99
9	0.39	0.38	2.93	0.26	2.27
10 (High)	0.96	0.58	3.48	0.51	2.98
Spread (Port. 10 – Port. 1)	2.03	0.31	3.16	0.59	3.55

Conclusion

- Contributions
  - Geopolitical Risk & Asset Pricing: Identifies US-China tension as a new priced risk factor independent of existing measures
  - Hedge Fund Performance: Shows that exposure differences to US-China tension explain cross-sectional return heterogeneity
  - Hedge Fund Timing Ability: Provides evidence that hedge funds can time US-China geopolitical risk
  - Overall Insight: Establishes US-China tension as a persistent and priced source of risk in financial markets
- Thanks for this opportunity to discuss this interesting paper!